# Practice Problems: Polynomials, Rational Functions, Long/Synthetic Division, and Solution of Rational Functions

Course: Pre-Calculus-I

AH Sheikh

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## Instructions

- Attempt every problem without using calculus.
- For rational equations and inequalities, always state the domain before clearing denominators or building a sign chart..
- For sketching, show domain, asymptotes, intercepts, holes, and a sign chart.

**Note 1.** *Problems are grouped as **Easy**, **Medium**, and **Challenging**. Each section contains five problems, for a total of fifteen.*

## Easy Problems

**E1.** State the degree, leading coefficient, and end behaviour of

$$P(x) = 4x^3 - 7x^2 + 2x - 9.$$

**E2.** Factor completely and find all roots:

$$P(x) = x^3 - 4x.$$

**E3.** Find the domain and all vertical asymptotes of

$$f(x) = \frac{5x - 2}{x^2 - 9}.$$

**E4.** Use synthetic division to divide

$$P(x) = 2x^3 - 5x^2 + 3x - 4$$

by $x - 2$. State the quotient and remainder.

**E5.** Solve the rational equation

$$\frac{3}{x} = \frac{2}{x + 1}.$$

State the domain first and check for extraneous roots.

## Medium Difficulty

**M6.** Divide $P(x) = x^4 - 3x^3 + 2x - 5$ by $D(x) = x^2 + 1$ using polynomial long division. Write the answer in the form $P(x) = D(x)Q(x) + R(x)$.

**M7.** Let $P(x) = x^4 - 2x^3 - 7x^2 + 8x + 12$.

(a) Use the Remainder Theorem to evaluate $P(-1)$ and $P(3)$.

(b) Based on your answers, determine whether $(x + 1)$ and $(x - 3)$ are factors of $P$.

**M8.** Analyse $f(x) = \dfrac{x^2 - 4}{x^2 - x - 6}$.

(a) Factor the numerator and denominator.

(b) State the domain and identify any holes.

(c) Find all vertical and horizontal asymptotes.

(d) Find the $x$- and $y$-intercepts.

**M9.** Solve the rational inequality

$$\frac{x - 2}{x + 4} \leq 0.$$

Show the sign chart and state the solution in interval notation.

**M10.** Solve the rational equation

$$\frac{1}{x - 1} + \frac{1}{x + 1} = \frac{4}{x^2 - 1}.$$

State the domain, then check for extraneous roots.

## Challenging Problems

**C11.** Let $f(x) = \dfrac{x^2 + x - 12}{x^2 - 2x - 3}$.

(a) Factor and simplify, stating the domain.

(b) Identify the hole, its coordinates, and any vertical asymptote.

(c) Find the horizontal asymptote and both intercepts.

(d) Sketch the graph, including the sign of $f$ on each interval determined by the asymptote, the hole, and the intercepts.

**C12.** Let $f(x) = \dfrac{2x^3 - 3x^2 - 8x + 12}{x^2 - 4}$.

(a) Use the Factor Theorem (via synthetic division) to show that $(x - 2)$ is a factor of the numerator.

(b) Fully factor the numerator, then simplify $f$.

(c) State the domain, and identify any holes with their coordinates.

(d) Find any vertical, horizontal, or oblique asymptote.

(e) Describe the end behaviour of $f$ as $x \to \pm\infty$.

**C13.** A rational function $f$ has:

- vertical asymptotes at $x = -2$ and $x = 1$,
- a horizontal asymptote $y = -3$,

- an $x$-intercept at $x = 4$,
- a hole at $x = 0$.

(a) Write a possible formula of the form

$$f(x) = \frac{a\,x\,(x-r)}{x\,(x+2)(x-1)}$$

and choose $a$ so the horizontal asymptote is $y = -3$.

(b) State the domain and the coordinates of the hole.

(c) Find the $y$-intercept of the simplified function.

**C14.** Solve the inequality

$$\frac{x^2-4}{x^2-2x-3} \geq 0.$$

Show a complete sign chart with all critical points (numerator zeros, denominator zeros), and state the solution in interval notation. Pay careful attention to whether endpoints are included.

**C15.** A tank is filled by three pipes:

- Pipe A fills the tank in 6 hours,
- Pipe B fills the tank in 9 hours,
- Pipe C drains the tank in 18 hours.

(a) Set up a rational equation for the time $t$ (in hours) it takes to fill the tank when all three are open simultaneously.

(b) Solve the equation, stating the domain, and interpret your answer.

(c) Verify the solution numerically by computing the fraction of the tank each pipe contributes at time $t$.

## Selected Hints

- **E3:** Factor $x^2 - 9 = (x-3)(x+3)$; vertical asymptotes occur where the denominator is zero and the numerator is nonzero.
- **E5:** Domain: $x \neq 0$ and $x \neq -1$. Cross-multiplying gives $3(x+1) = 2x$.
- **M6:** Insert zero coefficients: $x^4 - 3x^3 + 0x^2 + 2x - 5$.
- **M9:** Use the sign chart method; note that $x = -4$ is excluded (denominator zero) and $x = 2$ is included (numerator zero, non-strict inequality).
- **M10:** $x^2 - 1 = (x-1)(x+1)$; LCD is $(x-1)(x+1)$. Check whether either candidate is excluded.
- **C11:** Numerator factors as $(x+4)(x-3)$; denominator factors as $(x-3)(x+1)$. The factor $(x-3)$ cancels.
- **C12:** Synthetic division of the numerator by $x = 2$ gives remainder 0. Continue factoring the resulting quotient.
- **C13:** Set $a$ from the horizontal asymptote by comparing leading coefficients of the simplified form.

- **C14:** Critical points from numerator zeros ($x = \pm 2$) and denominator zeros ($x = -1, 3$). Use test points in each of five intervals.
- **C15:** Rate equation: $\dfrac{1}{6} + \dfrac{1}{9} - \dfrac{1}{18} = \dfrac{1}{t}$.

**Note 2.** *Attempt every problem on your own before checking the hints.*